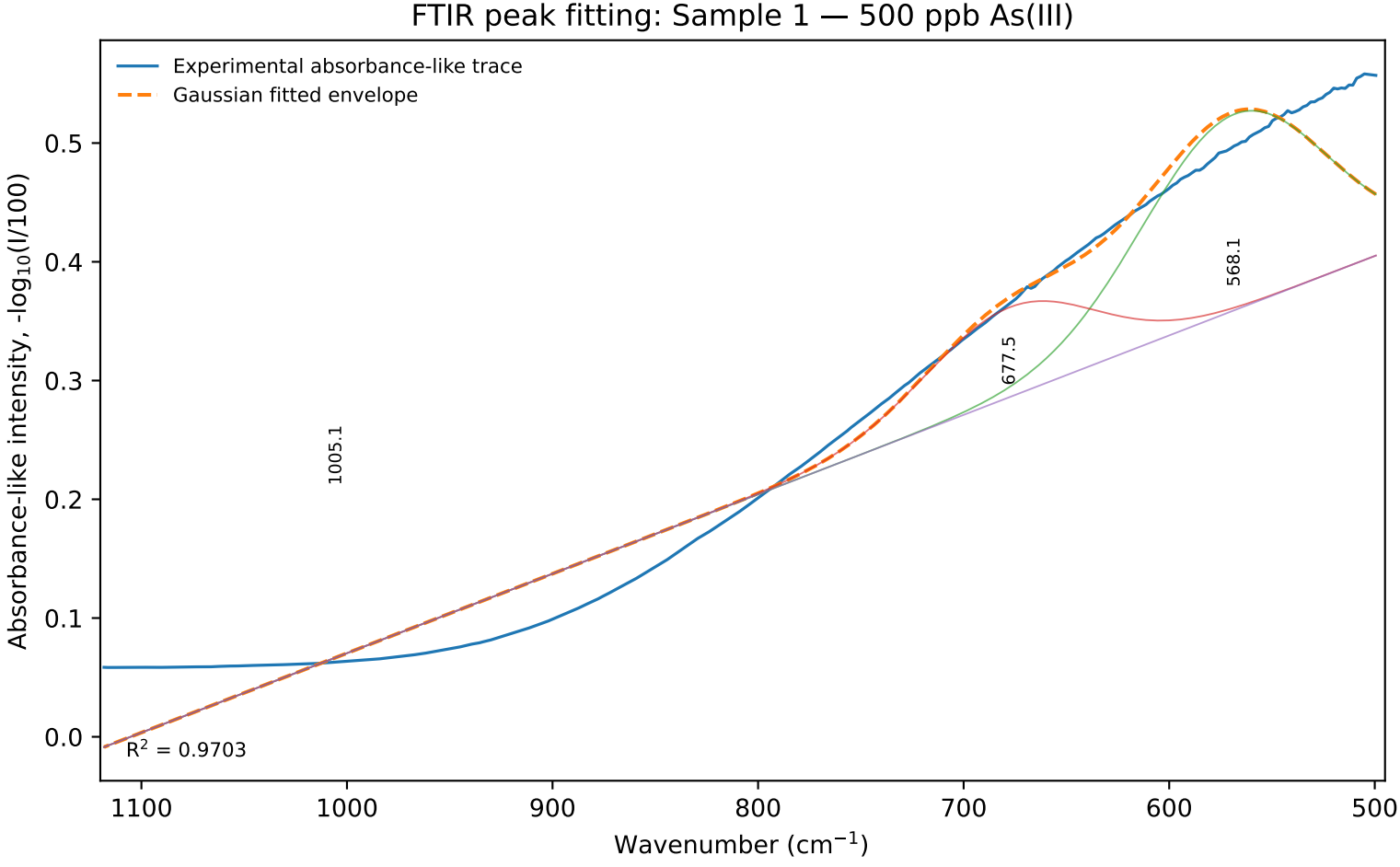
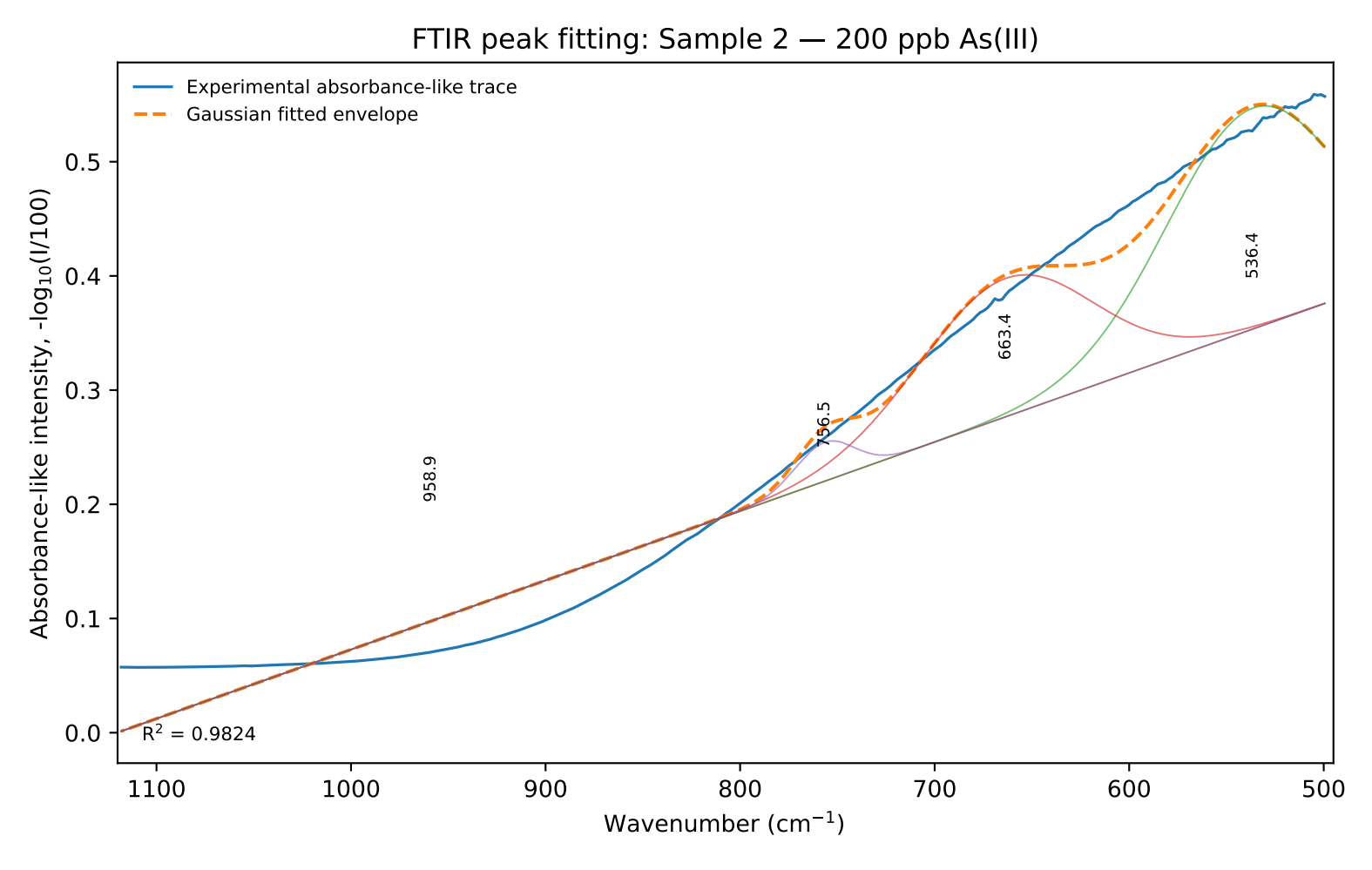
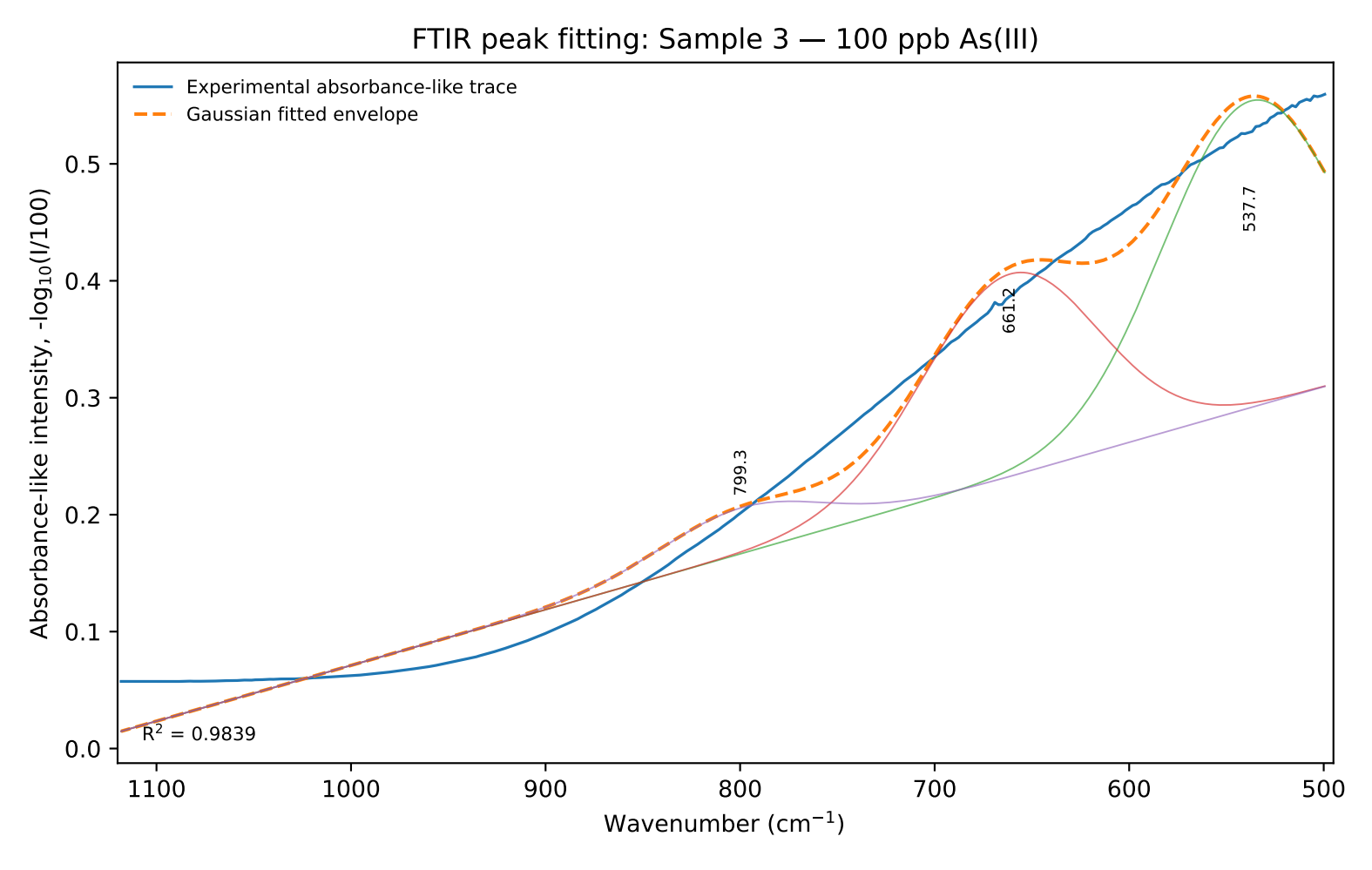
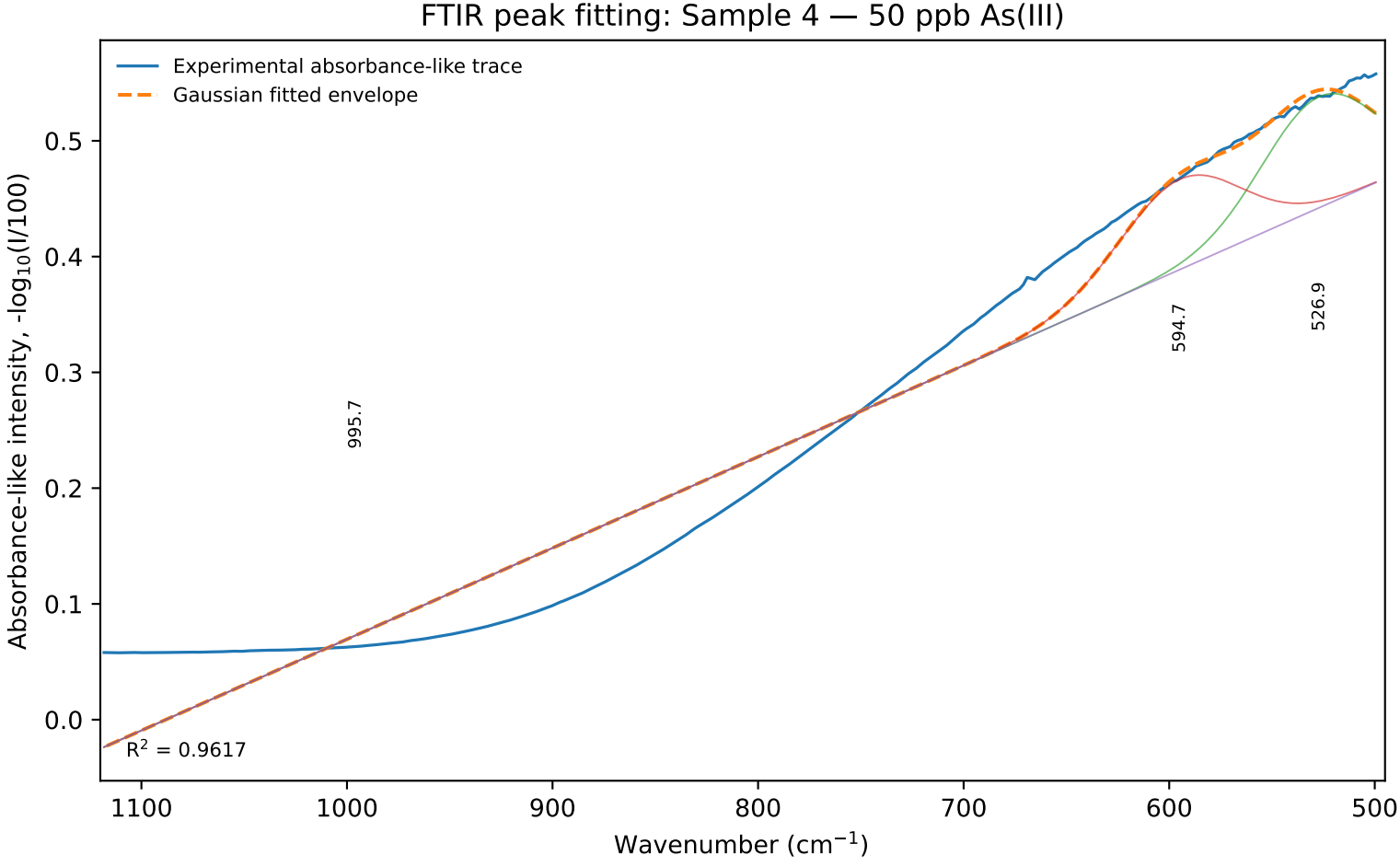
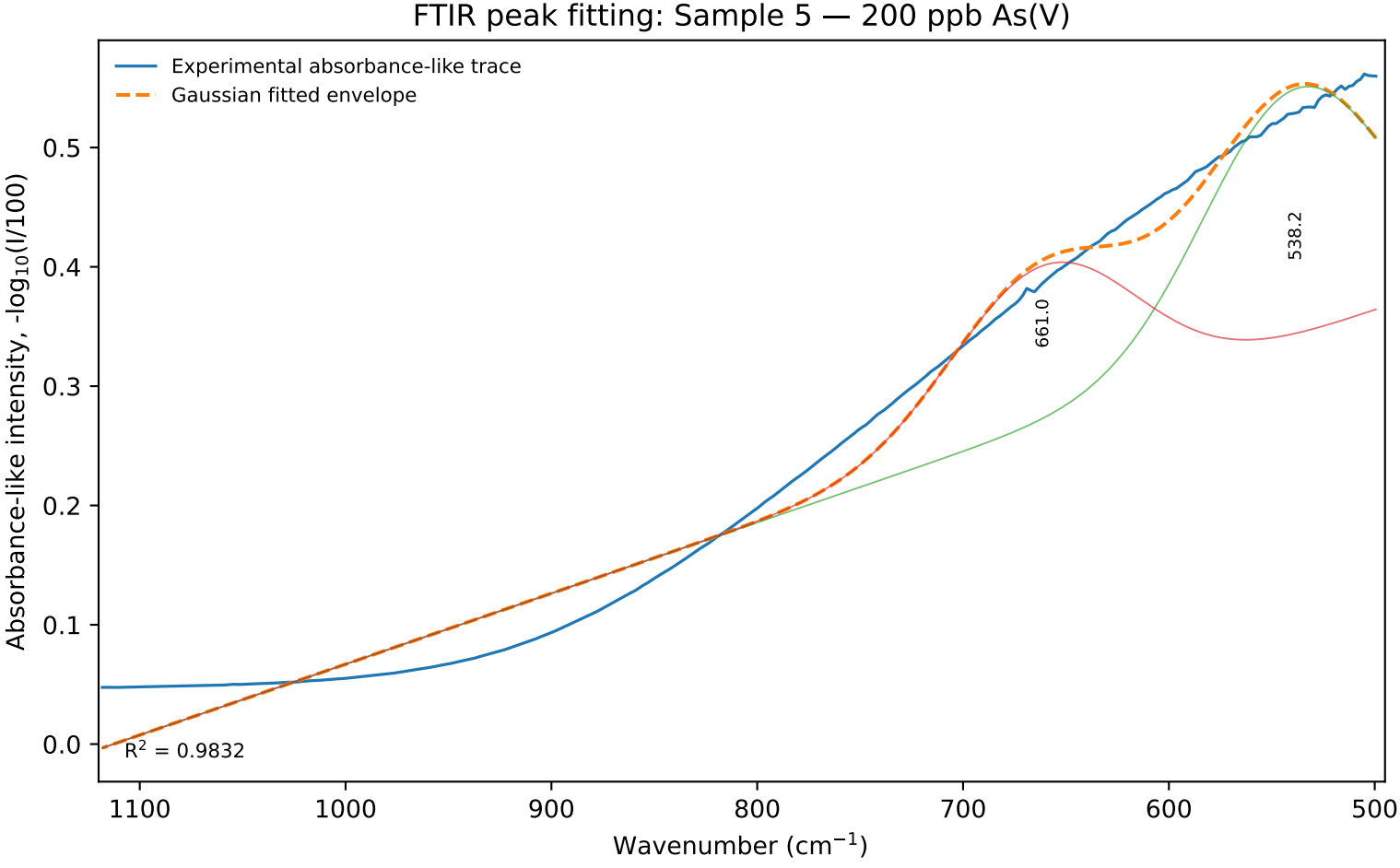
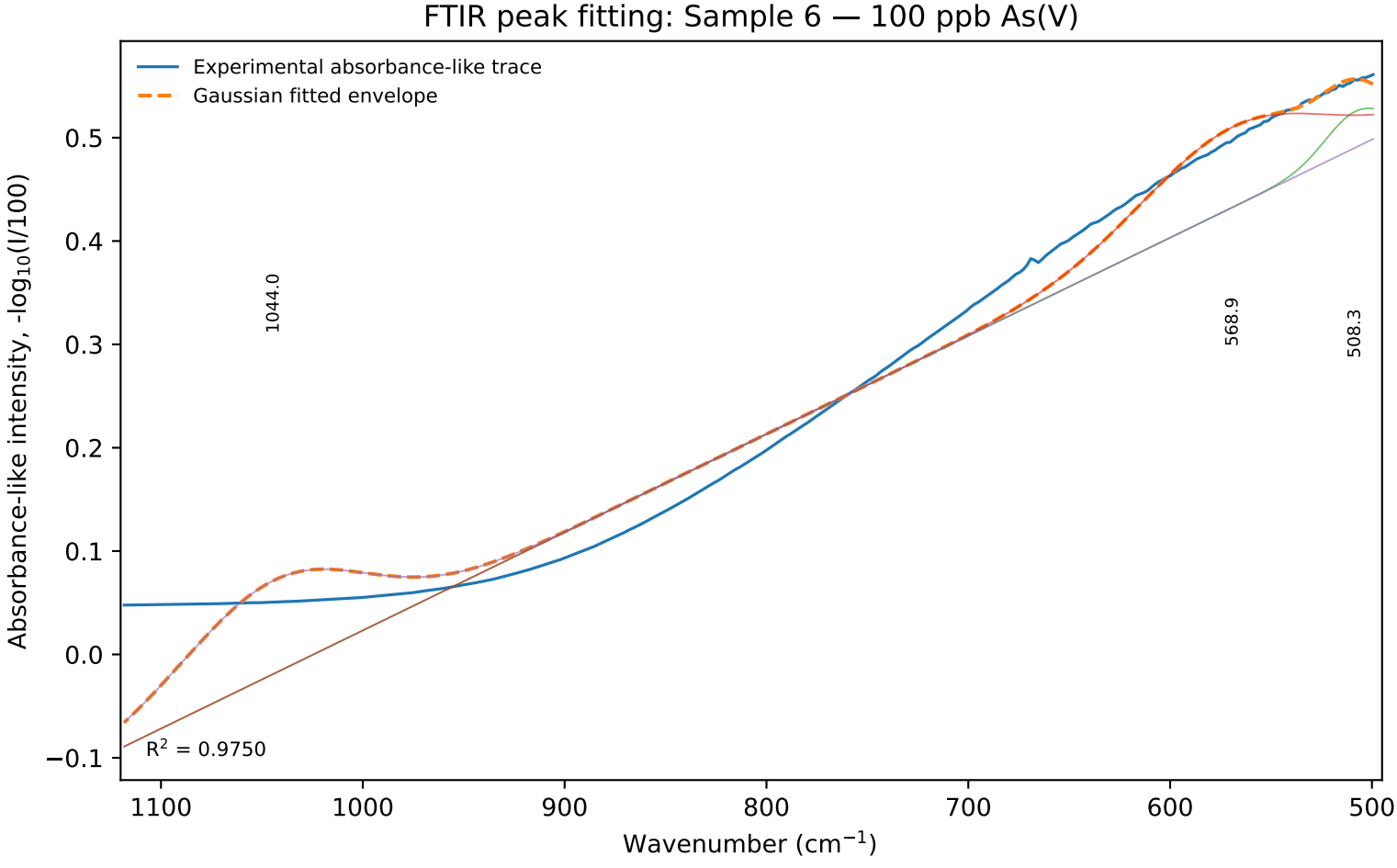

FTIR peak fitting: Sample 1 — 500 ppb As(III)
Experimental absorbance-like trace
Gaussian fitted envelope
1005.1
677.5
568.1
R² = 0.9703
Absorbance-like intensity, -log10(I/100)
Wavenumber (cm⁻¹)
0.0
0.1
0.2
0.3
0.4
0.5
1100
1000
900
800
700
600
500

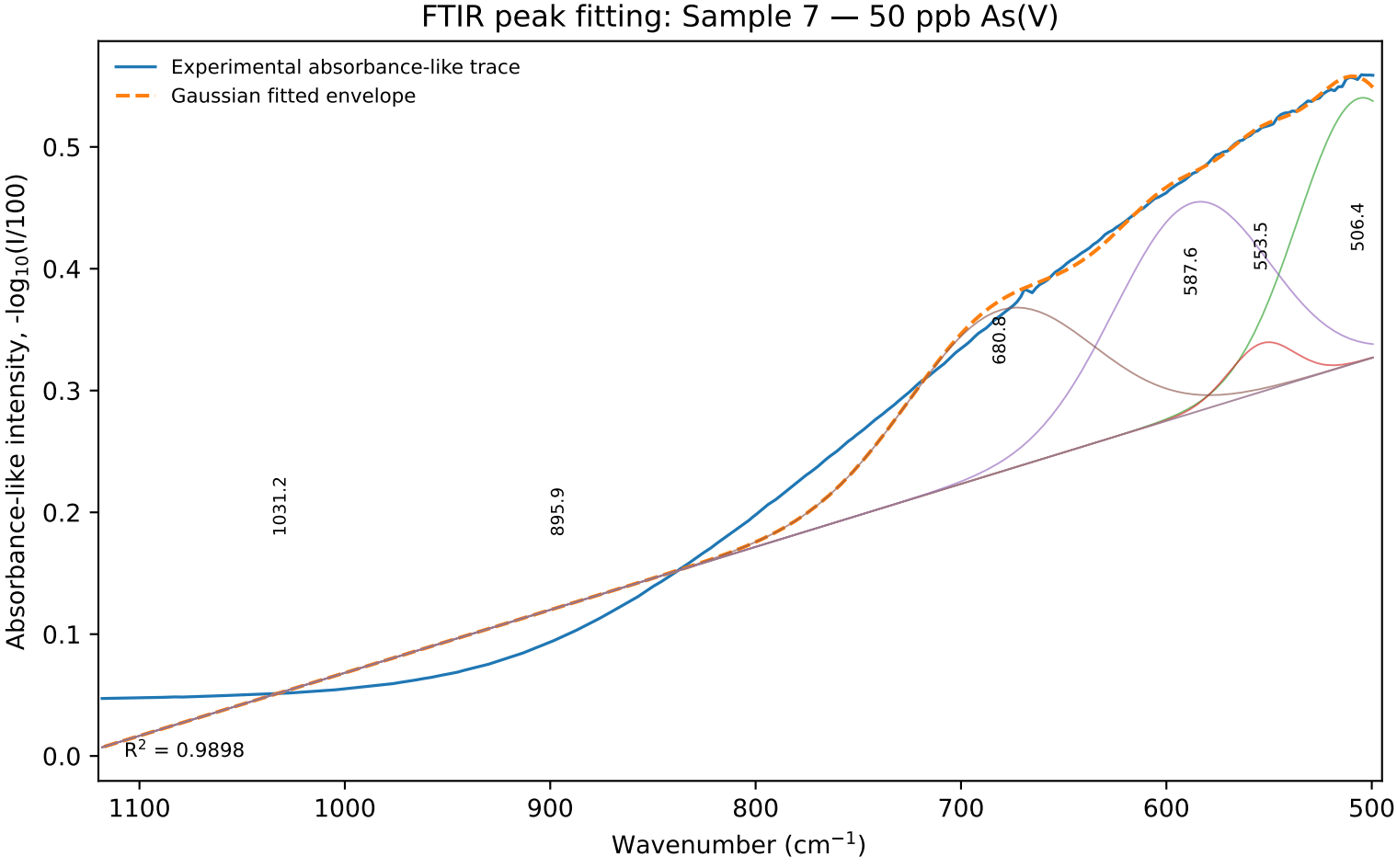

FTIR peak fitting: Sample 2 — 200 ppb As(III)
Experimental absorbance-like trace
Gaussian fitted envelope
958.9
756.5
663.4
536.4
R² = 0.9824
Absorbance-like intensity, -log10(I/100)
Wavenumber (cm⁻¹)
0.0
0.1
0.2
0.3
0.4
0.5
1100
1000
900
800
700
600
500

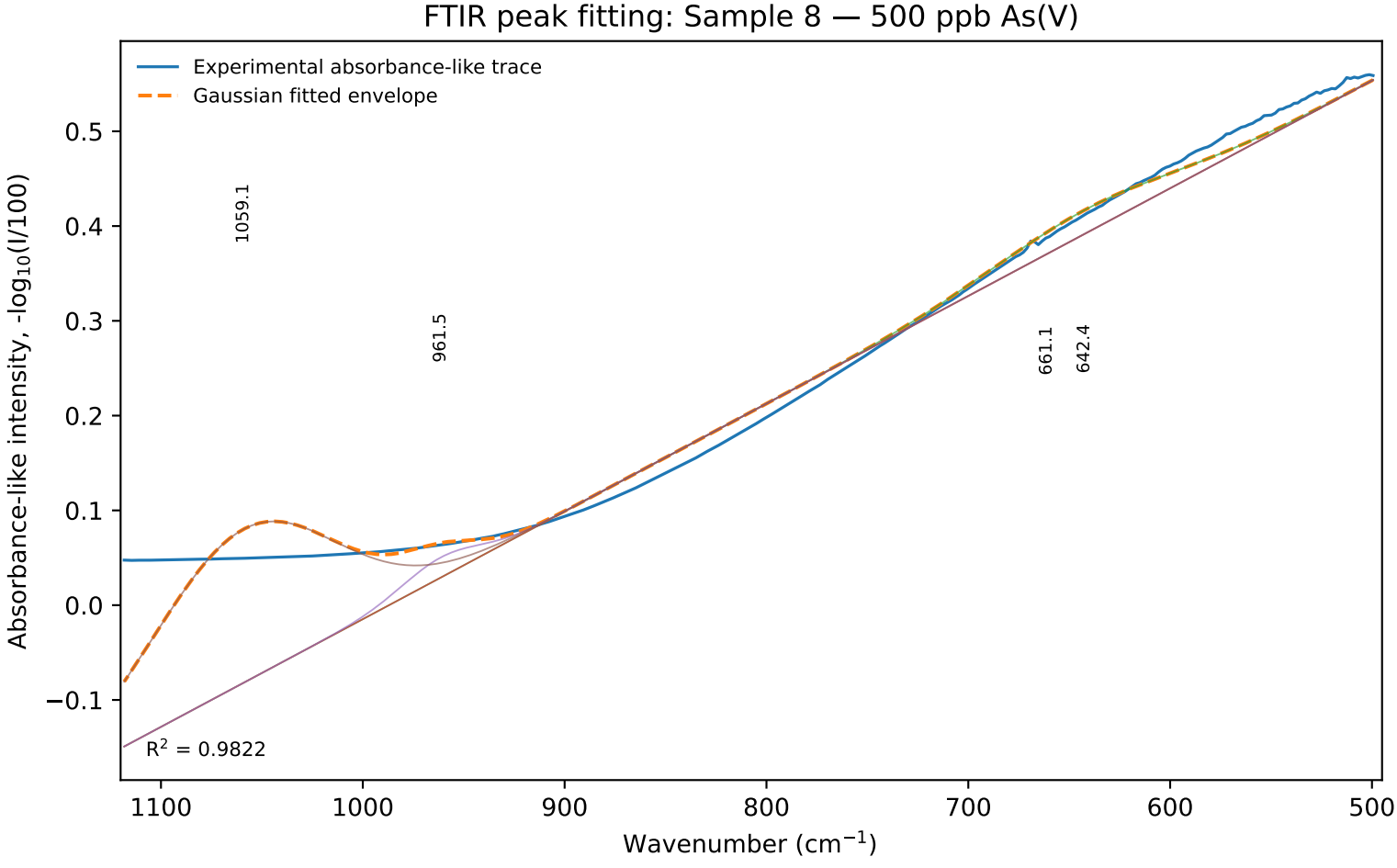

FTIR peak fitting: Sample 3 — 100 ppb As(III)
Experimental absorbance-like trace
Gaussian fitted envelope
799.3
661.2
537.7
R² = 0.9839
Absorbance-like intensity, -log10(I/100)
Wavenumber (cm⁻¹)
0.0
0.1
0.2
0.3
0.4
0.5
1100
1000
900
800
700
600
500

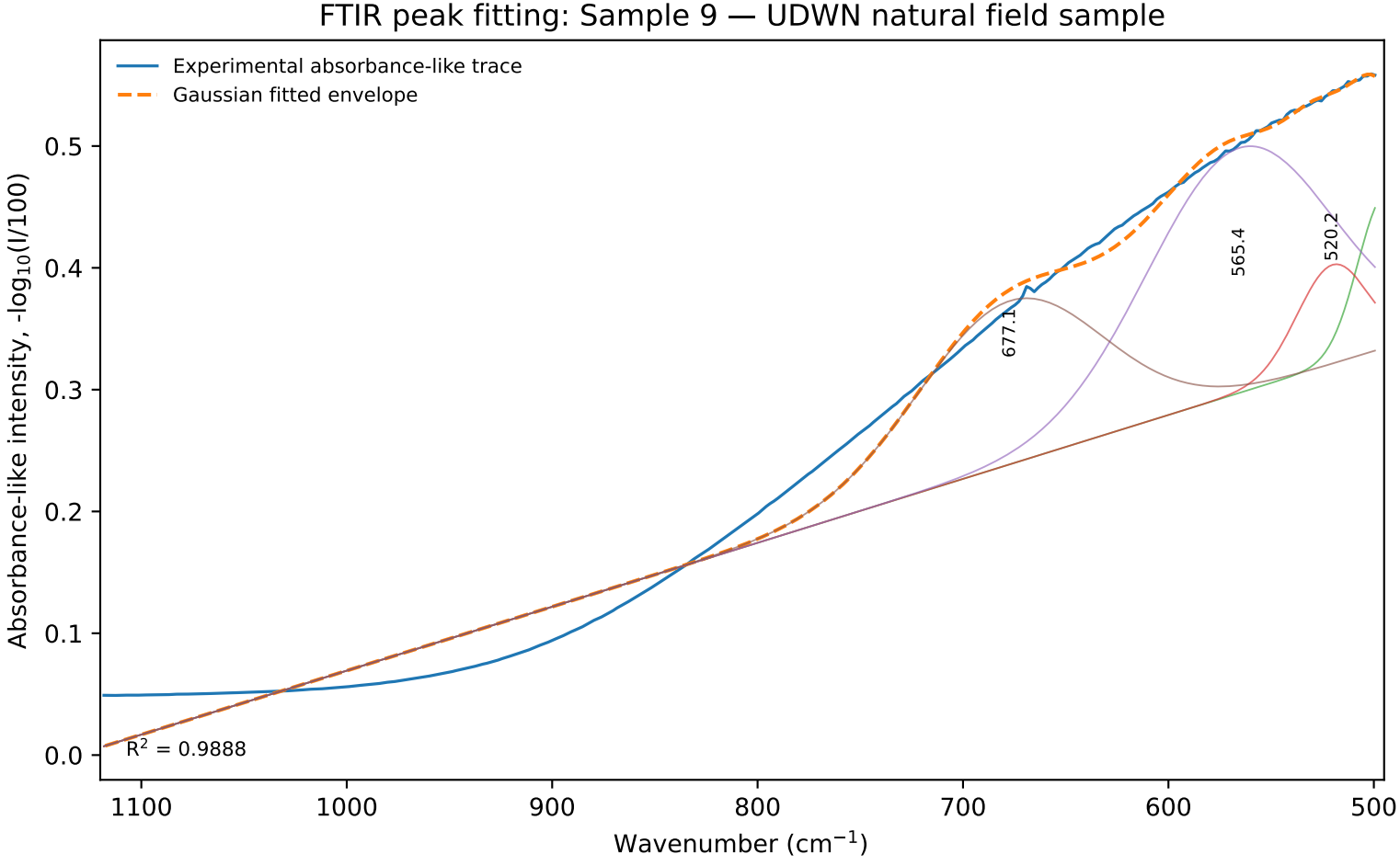

FTIR peak fitting: Sample 4 — 50 ppb As(III)
Experimental absorbance-like trace
Gaussian fitted envelope
995.7
594.7
526.9
R² = 0.9617
Absorbance-like intensity, -log10(I/100)
Wavenumber (cm⁻¹)
0.0
0.1
0.2
0.3
0.4
0.5
1100
1000
900
800
700
600
500

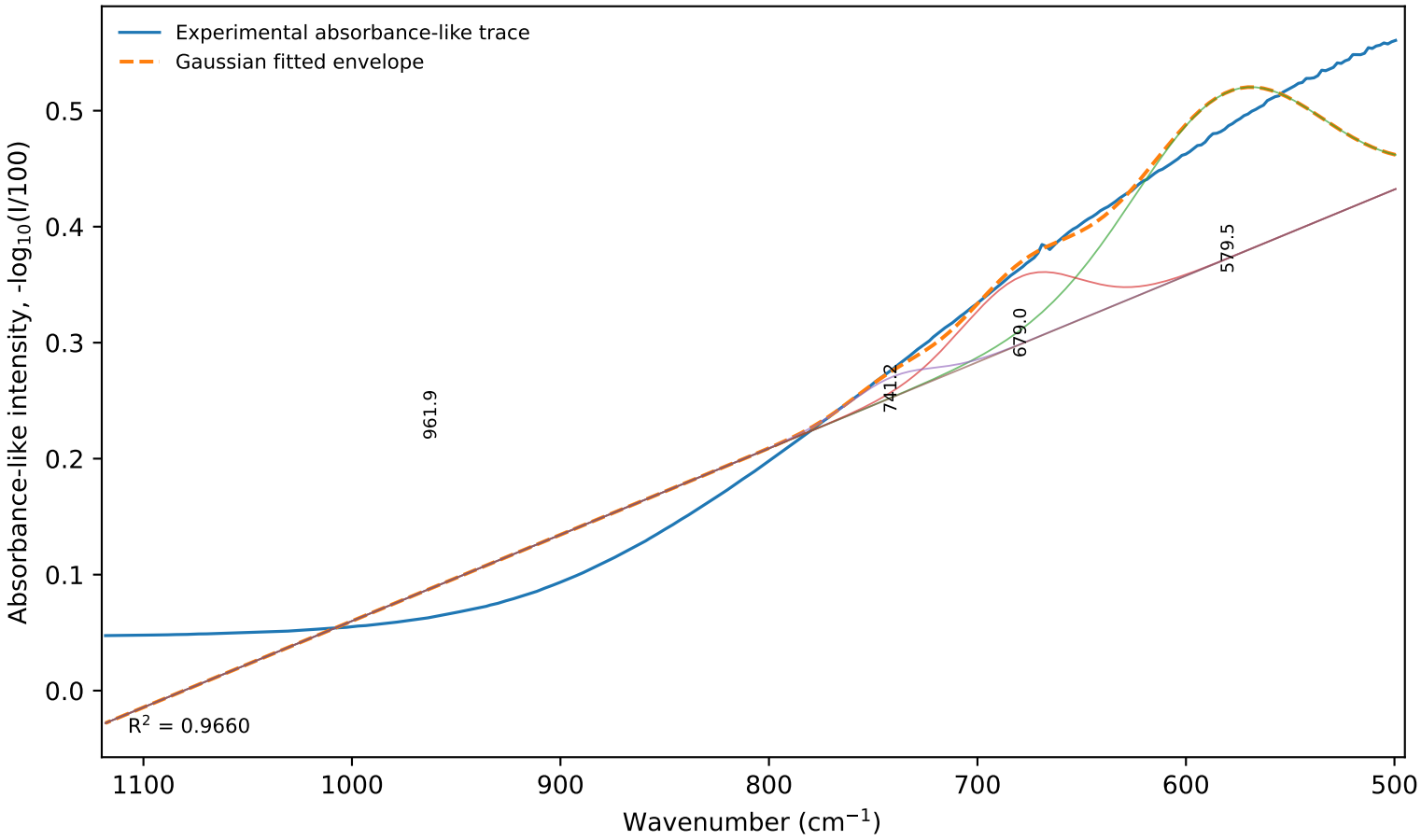

FTIR peak fitting: Sample 5 — 200 ppb As(V)
Experimental absorbance-like trace
Gaussian fitted envelope
661.0
538.2
R² = 0.9832
Absorbance-like intensity, -log₁₀(I/100)
Wavenumber (cm⁻¹)
0.0
0.1
0.2
0.3
0.4
0.5
1100
1000
900
800
700
600
500


FTIR peak fitting: Sample 6 — 100 ppb As(V)
Experimental absorbance-like trace
Gaussian fitted envelope
1044.0
568.9
508.3
R² = 0.9750
Absorbance-like intensity, -log10(I/100)
Wavenumber (cm⁻¹)
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
1100
1000
900
800
700
600
500


FTIR peak fitting: Sample 7 — 50 ppb As(V)
Experimental absorbance-like trace
Gaussian fitted envelope
1031.2
895.9
680.8
587.6
553.5
506.4
R² = 0.9898
Absorbance-like intensity, -log10(I/100)
Wavenumber (cm⁻¹)
0.0
0.1
0.2
0.3
0.4
0.5
1100
1000
900
800
700
600
500


FTIR peak fitting: Sample 8 — 500 ppb As(V)
Experimental absorbance-like trace
Gaussian fitted envelope
1059.1
961.5
661.1
642.4
R² = 0.9822
Absorbance-like intensity, -log₁₀(I/100)
Wavenumber (cm⁻¹)
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
1100
1000
900
800
700
600
500


FTIR peak fitting: Sample 9 — UDWN natural field sample
Experimental absorbance-like trace
Gaussian fitted envelope
677.1
565.4
520.2
R² = 0.9888
Absorbance-like intensity, -log10(I/100)
Wavenumber (cm⁻¹)
0.0
0.1
0.2
0.3
0.4
0.5
1100
1000
900
800
700
600
500


FTIR peak fitting: Sample 10 — UDWN acidified field sample
Experimental absorbance-like trace
Gaussian fitted envelope
961.9
741.2
679.0
579.5
R² = 0.9660
Absorbance-like intensity, -log₁₀(I/100)
Wavenumber (cm⁻¹)
0.0
0.1
0.2
0.3
0.4
0.5
1100
1000
900
800
700
600
500